Advancements and Challenges in Gated Recurrent Units (GRU) for Text Classification: A Systematic Literature Review

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***Abstract*— This systematic literature review (SLR) examines recent advancements and persisting challenges in using Gated Recurrent Units (GRU) for text classification. GRUs, a variant of recurrent neural networks, have become essential for processing sequential data due to their ability to handle temporal dependencies effectively. This review synthesizes findings from studies published between 2019 and 2024, highlighting the evolution of GRU applications across diverse contexts, such as sentiment analysis, natural language inference, and topic detection. We discuss how hybrid models integrating GRUs with technologies like CNNs, Transformers, and attention mechanisms have improved performance and adaptability. However, these advancements come with increased computational demands and complexities, such as overfitting, particularly in complex datasets. The SLR provides a comprehensive overview of internal modifications, hybrid approaches that enhance GRU models, and strategies to mitigate their limitations. Future directions are suggested for optimizing GRU efficiency, developing innovative architectures, and improving model interpretability to ensure ethical and effective deployment in real-time analytics and edge computing.**

Keywords*— GRU hybrid models, BiGRU, text classification, attention mechanism

I. INTRODUCTION

Gated Recurrent Units (GRU), a variant of Recurrent Neural Networks (RNNs), have made significant strides in the field of text classification due to their efficient handling of temporal dependencies within sequential data. GRUs were initially designed as a simpler alternative to traditional RNNs, yet they retain the capability to process sequences with long-term dependencies, making them particularly effective for complex text data tasks.

As the use of GRUs in text classification has increased, numerous studies have explored various modifications and hybrid approaches that integrate GRUs with other technologies such as Convolutional Neural Networks (CNNs), Transformers, and attention mechanisms to enhance performance and adaptability. These integrations have been shown to improve the model's ability to focus on relevant parts of the text, thereby increasing classification accuracy (Liu et al., 2020). For example, hybrid models that combine CNNs and GRUs leverage the strengths of both architectures by capturing local features through convolutional operations and sequential dependencies via recurrent units [1].

Despite these advancements, several challenges persist in the application of GRUs for text classification. Issues such as overfitting, high computational complexity, and difficulties in handling extremely long or noisy text sequences continue to impede optimal performance [2]. Additionally, the interpretability of GRU-based models remains a concern, as understanding the decision-making process of deep learning models is essential, particularly in critical applications such as healthcare and finance [3].

Therefore, this systematic literature review aims to critically analyze the advancements and challenges associated with GRU models in text classification tasks. We follow the PRISMA methodology [4] to ensure a rigorous approach in selecting and analyzing the literature, presenting findings that address four key research questions related to enhancements, performance, challenges, and future research directions in the use of GRUs for text classification. By compiling and synthesizing findings from various studies, this review aims to provide valuable insights for researchers and practitioners in the fields of Natural Language Processing (NLP) and machine learning, as well as to drive further innovations in the application of GRU models.

II. METHODS

A. Research Questions

This systematic literature review addresses the following research questions: (1) What are the primary enhancements and hybrid approaches implemented in GRU models for text classification? (2) How have various internal modifications and hybrid approaches enhanced the performance and efficiency of GRU architectures in text classification tasks? (3) What are the primary challenges in implementing GRU

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models for text classification, and how can these challenges be effectively addressed to enhance model performance and scalability? (4) What strategies can be implemented to address the challenges in GRU-based text classification systems?

B. Search Strategy

To identify relevant literature, a structured search strategy was employed:

1) Databases Used: Scopus was the primary database used for the search. It was chosen because it covers peer-reviewed science, technology, and social science literature comprehensively.

2) Keywords and Search Terms: The search terms included combinations of the following: (GRU OR BiGRU OR "Bi-GRU") AND (Classification OR Text OR Sentiment). These terms were selected to capture a wide range of studies focusing on applying GRU models in text classification and related tasks.

C. PRISMA

The PRISMA flow diagram below illustrates the study selection process for this systematic literature review. It details the number of records identified, screened, assessed for eligibility, and ultimately included in the final review.

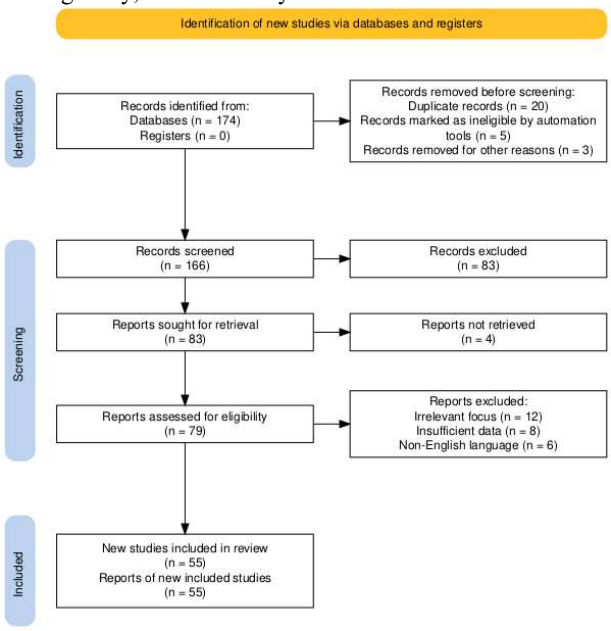


Fig. 1. The PRISMA flow diagram

III. Result

A. Reviewed Studies on Various Methods for GRU

Table 1 provides an overview of reviewed studies on various methods and their respective citations for GRU-based approaches from 2019 to 2024. Each year includes a range of techniques, encompassing both internal modifications and hybrid approaches, highlighting the evolution and diversification of GRU applications in natural language processing and data analysis.

TABLE I. Summary of GRU-Based Methods by Year

Year	Method
2019	GRU-CNN [5], BiGRU with multi-attention [6], Two-Channel CNN-GRU Fusion [7]
2020	GRU Cells [8], Multi-Channel CNN and BiGRU [1]S06, Double BiGRU and Knowledge Transfer [9], Bert-BiGRU-Softmax [10], BiGRU Based on Grammar Rules and BERT [11], CNN-BiGRU and Multihead Attention [2] , MI-biGRU [3], GRU-AI [12]
2021	CNN-BiGRU-AT [13], Multi-Channel CNN and BiGRU with Attention[14], Multi-Head Attention Capsules with Multi-Channel CNN and BiGRU [15], AT-BiGRU [16], GRU and TF-IDF [17], BiGRU-Dattention-DSSM [18] BiGRU with Pooling and Attention [19]
2022	BiGRU and Multi-Attention[20], Ensemble LSTM-GRU[21], CNN + BiGRU [22], BiGRU Language Model [23], BiLSTM-BiGRU and 1-D CNN [24], BiGRU Attention Capsule [25], GCNN with Self-Attention [26]
2023	BiGRU with BERT and Multi-Head Self-Attention [27], Multi BiGRU and Multi CNN [28], GRU and CNN with XLNet [29], Skip-GRU and Self-Attention [30] BiGRU with Attention [31] RoBERTa-GRU [32], Transformer-Encoder-GRU (T-E-GRU) [33], BiGRU and Improved Attention [34], TCN-BiGRU with Dual Attention [35], BERT-FGM-BiGRU [36], BERT-BiGRU [37], AB-GRU [38], XLNet-BiGRU-Att [39], Bi-LSTM and GRU with Poisson LDA [40], BiGRU and Graph Attention Network (BGAT) [41], Multichannel BiGRU [42], TCN-BiGRU [43], LSTM-GRU [44] BiGRU with Correlation Enhancement [45], BiGRU with Ernie-Tiny [46], CNN-BiGRU with Multi-Head Attention [47], CNN-BiGRU Fusion [48], CNN-BiGRU [49], Two-Channel CNN-GRU Fusion [48]
2024	Two-State GRU [50] GRU-CAP [51], Dual BiGRU-CNN [52], XLNet-CNN-GRU [53], Hybrid GRU with Word Embedding [54], XLNet-CNN-GRU [53] Hybrid GRU [54]

B. Enhancements and Hybrid Approaches in GRU Architectures

GRU is pivotal in processing sequential data when exploring deep learning architectures. This segment explores two critical facets: internal modifications and hybrid approaches to enhance GRU architectures. Table II provides an overview of various internal modifications implemented in GRU architectures to improve their performance.

TABLE II. Internal Modifications in GRU Architectures

No	Modification	Description	Primary Advantage	Ref.
1	Attention Mechanism	Highlights significant contributions from text segments.	Improved focus on relevant text information	[1], [13], [20], [27]
2	Grammar Rules	Integrates grammar rules with GRU to handle complex syntactic structures.	Enhanced handling of grammar complexity	[11]
3	Pooling	Pooling methods are used to summarize important information from text efficiently.	Efficient information summarization	[19]
4	Capsule Network	Combines GRU with Capsule Networks to capture richer features.	Richer feature extraction	[9], [25]
5	FGM (Fast Gradient Method)	Employs adversarial training to improve model generalization by introducing perturbations.	Improved model generalization	[36]
6	Correlation Enhancement	Enhances cross-modal feature correlations for sentiment analysis.	Improved feature correlation	[45]
7	Position Features	It uses position features to enrich word embeddings	Enhanced word embeddings	[3]

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No	Modification	Description	Primary Advantage	Ref.
		and improve accuracy.		
8	Multi-Task Learning	Transfers knowledge from short text review classification to drug review classification tasks.	Improved performance through knowledge transfer	[9]
9	Knowledge Transfer	Transfers knowledge from one task to another to enhance model performance.	Improved task performance	[9]
10	Embedding Optimization	Optimizes word embeddings to improve semantic representation in the model.	Enhanced semantic representation	[32], [54]
11	BiGRU (Bidirectional GRU)	Utilizes bidirectional GRU to capture context from both directions.	Comprehensive context understanding	[1], [11], [13], [23], [27],
2	Double GRU	It uses two layers of GRU to capture complex features.	Improved feature capture capacity	[9], [25]

Table III examines the integration of GRUs with other neural network architectures, highlighting how these combinations enhance data processing capabilities. It showcases various hybrid models, their advantages, and their applications, illustrating the versatility and potential of GRUs in modern computational tasks.

TABLE III. HYBRID METHOD IN GRU ARCHITECTURES

Hybrid Method	Primary Advantage	Ref.
GRU + CapsNet	Captures richer features for sentiment analysis.	[42]
GRU + CNN	Enhances feature extraction and contextual understanding.	[5], [7], [55]
GRU + CNN, XLNet	Leverages robust embeddings and feature extraction.	[29]
GRU + LSTM	Improves model accuracy and robustness through ensemble methods.	[12], [33], [36]
GRU + RoBERTa	Provides robust contextual embeddings and sequence modeling.	[22]
GRU + TF-IDF	Enhances sentiment classification by leveraging word importance.	[17]
BiGRU + BERT	Enhances text representation and label correlation handling.	[10], [11], [27], [37]
BiGRU + CNN	Combines local feature extraction with sequence modeling for improved accuracy.	[1], [2], [13], [22]
BiGRU + DSSM	Enhances long text feature extraction and matching accuracy.	[18]
BiGRU + Ernie-Tiny	Improves sentiment analysis with visualized interaction.	[46]
BiGRU + Graph Attention Network	Captures dependencies between aspect words and sentiment words.	[41]
BiGRU + Language Model	Provides dynamic sentiment representation based on context.	[23]
BiGRU + LSTM, CNN	Integrates multiple layers for comprehensive feature extraction.	[24]
BiGRU + TCN	Combines temporal convolution with GRU for enhanced sentiment analysis.	[35], [43]
BiGRU + XLNet	Leverages advanced transformer models for emotion detection.	[39]

C. Effectiveness in Various Contexts

In the diverse landscape of deep learning applications, Table IV offers a comprehensive overview of how GRU enhances functionality across different domains when combined with other neural network architectures. This table categorizes various hybrid GRU models and delineates their effectiveness in specific applications such as natural language processing, image captioning, sentiment analysis, etc. Each model is evaluated on its ability to integrate and process multiple data types, improve contextual understanding, and increase computational efficiency, showcasing the broad utility and adaptability of GRU-based solutions in addressing complex and varied computational challenges.

TABLE IV. PERFORMANCE ENHANCEMENTS AND APPLICATIONS OF GRU IN TEXT CLASSIFICATION

Metode	Context	Performance Metrics	Ref.
Bert-BiGRU-Softmax	Sentiment Analysis in E-commerce	Accuracy: >95.5%	[10]
BiGRU with Improved Attention	Fault Text Classification in Railway Systems	Accuracy: 95.6%, Precision: 94.7%, Recall: 95.2%, F1-Score: 94.9%	[34]
BiGRU with Pooling and Attention	Text Classification in NLP	Precision: +5.9%, Recall: +5.8%, F1-Score: +4.6%	[19]
BiLSTM-BiGRU and 1-D CNN	Binary Sentiment Classification of Movie Reviews	Accuracy: 93.89%, F1-Score: 0.9828	[24]
CNN + BiGRU	Aspect-Based Sentiment Analysis	Accuracy improvement over baseline: 12.12%	[22]
CNN-BiGRU-AT	Sentiment Analysis in Social Media	Recall: >90%, F1-Score: >80%	[13]
Ensemble LSTM-GRU	Sentiment and Emotion Detection in Finance	Accuracy: 99%, F1-Score: 92%	[21]
GRU and CNN with XLNet	Sentiment Analysis on French Customer Reviews	Accuracy: 96.5%	[29]
RoBERTa-GRU	Sentiment Analysis across Multiple Datasets	Average Accuracy: 91.91%	[31]
T-E-GRU	Chinese Sentiment Analysis	Accuracy: 90.08%, F1-Score: 90.32%	[33]
TCN-BiGRU	PM2.5 Concentration Prediction	MAE: 4.20, RMSE: 7.71, R^2: 0.961	[43]
Two-Channel CNN-GRU Fusion	Stance Detection in Microblog Texts	Accuracy: 86.5%, F1-Score: 85.7%	[56]

D. Challenges Faced in the Implementation of GRU

Table V outlines the challenges of using GRU models in natural language processing and other machine-learning applications. These challenges include overfitting, where models may not generalize well to new data, high computational complexity, and difficulties handling long sequences, leading to vanishing gradients. The table also highlights challenges such as data sparsity and integrating GRU with other model architectures, emphasizing the complexity and considerations needed to optimize GRU performance.

TABLE V. CHALLENGES IN USING GRU MODELS

No.	Challenge	Description	Ref.
1	Overfitting	GRU models can memorize training data, leading to poor generalization.	[30], [40]

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No.	Challenge	Description	Ref.
2	Computational Complexity	GRUs are resource-intensive, requiring significant processing power.	[11], [21]
3	Handling Long Sequences	GRUs may face issues like vanishing gradients with very long sequences.	[1], [39]
4	Data Sparsity	Sparse data can degrade the performance of GRUs.	[44]
5	Limited Interpretability	GRUs, like many neural networks, are often not easily interpretable.	[21], [32]
6	Hyperparameter Tuning	Optimizing GRUs involves complex and time-consuming parameter tuning.	[32]
7	Scalability	Scaling GRUs for large datasets can be challenging due to computational demands.	[12], [51],
8	Sensitivity to Initialization	GRUs' performance can be significantly influenced by initial parameter settings.	[15], [34]
9	Integration with Other Models	Merging GRUs with other model types can require detailed architectural planning.	[10]
10	Real-time Processing	Achieving real-time processing with GRUs can be difficult in resource-limited settings.	[49], [50]

IV. Discussion

A. Summary of Key Findings

The utilization of Gated Recurrent Units (GRUs) in text classification has been a topic of extensive research, demonstrating substantial advancements and addressing various challenges inherent to natural language processing tasks. Table VI compiles critical insights from numerous studies, presenting an overview of how GRUs enhance performance, adapt across diverse domains, and their computational demands. This compilation also highlights the strategies employed to mitigate overfitting and integrate attention mechanisms to improve model interpretability. The referenced studies underscore the evolving landscape of deep learning techniques in handling textual data, offering a comprehensive view of the current capabilities and limitations of GRU-based models.

TABLE VI. Key Findings on the Use of GRUs in Text Classification

Aspect	Key Finding	Ref.
Performance Enhancement	Hybrid models like CNN-GRU and BERT-BiGRU excel in managing both spatial and sequential data, significantly boosting performance through improved feature extraction.	[7], [10], [36]
Adaptability Across Domains	GRUs require fine-tuning when transitioning between linguistic properties or data modalities to maintain effectiveness across various applications.	[31], [47], [55]
Training and Computational Demands	Due to the high complexity of training GRU models, models combining GRUs with BERT or Transformers demand increased training resources and hardware capabilities.	[33], [53]
Overfitting Challenges	Regularization strategies like dropout are essential to prevent overfitting and enhance generalization, as GRUs are prone to overfitting on limited datasets.	[2], [29], [41]
Model Interpretability	Efforts to incorporate attention mechanisms help improve focus and interpretability by elucidating influential data features addressing the complexity of GRU models.	[9], [19], [38]

B. Summary of Weaknesses in Various GRU-Based Methods for Sentiment Analysis and Text Classification

In exploring the effectiveness of various GRU-based methods in text classification, it's critical also to consider their limitations, as detailed in Table VII. This table comprehensively summarizes the weaknesses of prominent GRU-based models such as Bert-BiGRU-Softmax, BiGRU with Improved Attention, and others. These weaknesses range from high computational demands to generalizing across text data types. This analysis not only aids in understanding the potential pitfalls of each method but also highlights the importance of continued innovation and testing in developing more robust GRU-based classification systems. By examining these limitations, researchers and developers can better strategize improvements and optimize their models for enhanced performance and adaptability across various text analysis applications.

TABLE VII. Summary of Weaknesses in Various GRU-Based Methods for Sentiment Analysis and Text Classification

Method	Weaknesses	Ref.
Bert-BiGRU-Softmax	Requires a large amount of training data to perform well, especially on large datasets.	[10]
BiGRU with Improved Attention	Prone to overfitting on small datasets.	[34]
BiGRU with Pooling and Attention	Faces challenges in handling very long or very short texts.	[19]
BiLSTM-BiGRU and 1-D CNN	High model complexity leads to slower training and requires more resources.	[24]
CNN + BiGRU	Difficulty in handling very long contexts due to limitations in capturing long-term dependencies.	[22]
CNN-BiGRU-AT	Performance may degrade on very large and highly diverse datasets.	[13]
Ensemble LSTM-GRU	Requires a lot of training time and large computational resources.	[21]
GRU and CNN with XLNet	Prone to overfitting if the dataset is not large and diverse enough.	[29]
RoBERTa-GRU	It is computationally intensive and requires a lot of memory for training.	[32]
T-E-GRU	Difficulty in effectively handling very long or very short texts.	[33]
TCN-BiGRU	Prone to overfitting on small datasets.	[43]
Two-Channel CNN-GRU Fusion	High model complexity leads to slower training.	[7]

C. Implications for Practice

Advancements in GRU technologies have broad implications for various fields, from natural language processing to large-scale deployments. These improvements necessitate efficient integration and management strategies across performance, adaptability, and scalability. Table VIII outlines detailed implications for practice, highlighting how GRU models can be optimized for real-world applications.

TABLE VIII. Practical Implications of Advancements in GRU Architectures

No	Aspect	Implications for Practice	Ref.
1	Attention Mechanisms	Enhances GRU's focus on relevant input, improving performance in NLP tasks.	[1], [5], [6]
2	Hybrid Models with CNN	Improves feature extraction and classification accuracy, which is beneficial for image and text processing.	[1], [2], [5]

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No	Aspect	Implications for Practice	Ref.
3	Integration with Pretrained Models	Improves understanding and generation of language, helpful in chatbots and automated translation.	[5], [11]
4	Specificity-Sensitive Applications	Suitable for precise classification tasks such as medical diagnosis and niche sentiment analysis.	[8]
5	Multi-Attention Mechanisms	Enhances ability to handle complex structures, benefiting document summarization and question answering.	[1], [6], [15]
6	Overfitting Mitigation Techniques	Leads to more reliable GRU models for diverse datasets, useful in fraud detection.	[19], [26], [31]
7	Computational Efficiency	Results in efficient models suitable for real-time applications like speech recognition and live translation.	[5], [27], [55]
8	Data Augmentation Strategies	Improves robustness of GRU models, adaptable to varying data conditions in healthcare and finance.	[19], [24], [38]
9	Cross-Modal and Multimodal Sentiment Analysis	It enhances context understanding and benefits social media monitoring and customer feedback analysis.	[20], [38], [45]
10	Temporal Feature Extraction for Time Series Predictions	Leads to accurate predictions in financial forecasting, weather prediction, and other temporal data applications.	[34], [43], [44]

D. Potential Impact on Future Research and Applications

The integration and enhancement of GRUs in text classification tasks have shown significant potential in advancing research and broadening application scopes. These improvements can transform text analysis aspects, ensuring more precise and efficient processing across different languages and real-time applications. The following table, Table IX, outlines key advancements in GRU technology, highlighting their implications for future research and practical applications in text classification. These enhancements promise better performance in standard tasks and extend GRU applicability to more complex and dynamic text analysis scenarios.

TABLE IX. IMPACT OF GRU ADVANCEMENTS ON TEXT CLASSIFICATION RESEARCH AND APPLICATIONS

No.	Aspect	Impact	Ref.
1	Enhanced Feature Extraction	GRU advancements could enable a more nuanced understanding of textual nuances, improving accuracy in complex text classification tasks.	[30], [31]
2	Hybrid Models Efficiency	Integration of GRUs with other neural models like CNNs and Transformers will likely enhance text classification efficiency and accuracy.	[26], [27]
3	Multilingual Processing	Improved GRUs could better handle multiple languages without substantial changes to the architecture, broadening the scope of text classification applications globally.	[28], [29]
4	Real-time Text Analysis	Faster and more efficient GRUs could allow for real-time text analysis, which would be helpful in applications like live sentiment analysis or on-the-fly content moderation.	[32], [34]
5	Robustness Against Overfitting	Advancements in GRU designs may provide better mechanisms to combat overfitting, making models more generalizable across diverse text datasets.	[26], [27]

V. CONCLUSION

The exploration of Gated Recurrent Unit (GRU) architectures has significantly impacted deep learning, especially in processing sequential data for applications such as natural language processing and time series analysis. GRUs, enhanced with hybrid technologies like CNNs, Transformers, and attention mechanisms, show improved performance and efficiency. However, they face challenges such as high computational demands and a propensity for overfitting in complex scenarios. Future research should optimize GRU efficiency, develop innovative hybrid architectures, and enhance interpretability to ensure ethical use, especially in real-time analytics and edge computing. Addressing these issues is essential for advancing GRU's applicability and ensuring sustainable integration across various domains.

REFERENCES

[1] Y. Cheng, L. Yao, G. Xiang, G. Zhang, T. Tang, and L. Zhong, "Text Sentiment Orientation Analysis Based on Multi-Channel CNN and Bidirectional GRU With Attention Mechanism," IEEE Access, vol. 8, pp. 134964–134975, 2020, doi: 10.1109/ACCESS.2020.3005823.

[2] H. Qiu, C. Fan, J. Yao, and X. Ye, "Chinese Microblog Sentiment Detection Based on CNN-BiGRU and Multihead Attention Mechanism," Sci. Program., vol. 2020, pp. 1–13, Oct. 2020, doi: 10.1155/2020/8865983.

[3] X. Wang, X. Chen, M. Tang, T. Yang, and Z. Wang, "Aspect-Level Sentiment Analysis Based on Position Features Using Multilevel Interactive Bidirectional GRU and Attention Mechanism," Discret. Dyn. Nat. Soc., vol. 2020, pp. 1–13, Jul. 2020, doi: 10.1155/2020/5824873.

[4] N. R. Haddaway, M. J. Page, C. C. Pritchard, and L. A. McGuinness, "PRISMA2020: An R package and Shiny app for producing PRISMA 2020-compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis," Campbell Syst. Rev., vol. 18, no. 2, p. e1230, Jun. 2022, doi: https://doi.org/10.1002/cl2.1230.

[5] L. Luo, "Network text sentiment analysis method combining LDA text representation and GRU-CNN," Pers. Ubiquitous Comput., vol. 23, no. 3–4, pp. 405–412, Jul. 2019, doi: 10.1007/s00779-018-1183-9.

[6] J. Zhang, F. Liu, W. Xu, and H. Yu, "Feature Fusion Text Classification Model Combining CNN and BiGRU with Multi-Attention Mechanism," Futur. Internet, vol. 11, no. 11, p. 237, Nov. 2019, doi: 10.3390/fi11110237.

[7] W. Li, Y. Xu, and G. Wang, "Stance Detection of Microblog Text Based on Two-Channel CNN-GRU Fusion Network," IEEE Access, vol. 7, pp. 145944–145952, 2019, doi: 10.1109/ACCESS.2019.2944136.

[8] N. Gruber and A. Jockisch, "Are GRU Cells More Specific and LSTM Cells More Sensitive in Motive Classification of Text?," Front. Artif. Intell., vol. 3, Jun. 2020, doi: 10.3389/frai.2020.00040.

[9] Y. Han, M. Liu, and W. Jing, "Aspect-Level Drug Reviews Sentiment Analysis Based on Double BiGRU and Knowledge Transfer," IEEE Access, vol. 8, pp. 21314–21325, 2020, doi: 10.1109/ACCESS.2020.2969473.

[10] Y. Liu, J. Lu, J. Yang, and F. Mao, "Sentiment analysis for e-commerce product reviews by deep learning model of Bert-BiGRU-Softmax," Math. Biosci. Eng., vol. 17, no. 6, pp. 7819–7837, 2020, doi: 10.3934/mbe.2020398.

[11] Q. Lu, Z. Zhu, F. Xu, D. Zhang, W. Wu, and Q. Guo, "Bi-GRU Sentiment Classification for Chinese Based on Grammar Rules and BERT," Int. J. Comput. Intell. Syst., vol. 13, no. 1, p. 538, 2020, doi: 10.2991/ijcis.d.200423.001.

[12] W. Gu, S. Zheng, R. Wang, and C. Dong, "Forecasting Realized Volatility Based on Sentiment Index and GRU Model," J. Adv. Comput. Intell. Intell. Informatics, vol. 24, no. 3, pp. 299–306, May 2020, doi: 10.20965/jaciii.2020.p0299.

[13] W. Yan, L. Zhou, Z. Qian, L. Xiao, and H. Zhu, "Sentiment Analysis of Student Texts Using the CNN-BiGRU-AT Model," Sci. Program., vol. 2021, pp. 1–9, Oct. 2021, doi: 10.1155/2021/8405623.

[14] W. Ali, Y. Yang, X. Qiu, Y. Ke, and Y. Wang, "Aspect-Level Sentiment Analysis Based on Bidirectional-GRU in SIoT," IEEE

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Access, vol. 9, pp. 69938–69950, 2021, doi: 10.1109/ACCESS.2021.3078114.

[15] Y. Cheng et al., "Sentiment Analysis Using Multi-Head Attention Capsules With Multi-Channel CNN and Bidirectional GRU," IEEE Access, vol. 9, pp. 60383–60395, 2021, doi: 10.1109/ACCESS.2021.3073988.

[16] X. Lu and H. Zhang, "Sentiment Analysis Method of Network Text Based on Improved AT-BiGRU Model," Sci. Program., vol. 2021, pp. 1–11, May 2021, doi: 10.1155/2021/6669664.

[17] H. Yu, Y. Ji, and Q. Li, "Student sentiment classification model based on GRU neural network and TF-IDF algorithm," J. Intell. Fuzzy Syst., vol. 40, no. 2, pp. 2301–2311, Feb. 2021, doi: 10.3233/JIFS-189227.

[18] S. Chen and T. Xu, "Long Text QA Matching Model Based on BiGRU–DAttention–DSSM," Mathematics, vol. 9, no. 10, p. 1129, May 2021, doi: 10.3390/math9101129.

[19] Y. L. Hu and Q. S. Zhao, "Bi-GRU Model Based on Pooling and Attention for Text Classification," Int. J. Wirel. Mob. Comput., vol. 21, no. 1, p. 26, 2021, doi: 10.1504/IJWMC.2021.119057.

[20] M. M.Abdelgwad, "Arabic Aspect Based Sentiment Analysis Using Bidirectional GRU Based Models," J. King Saud Univ. - Comput. Inf. Sci., vol. 34, no. 9, pp. 6652–6662, 2022, doi: 10.1016/j.jksuci.2021.08.030.

[21] N. Aslam, F. Rustam, E. Lee, P. B. Washington, and I. Ashraf, "Sentiment Analysis and Emotion Detection on Cryptocurrency Related Tweets Using Ensemble LSTM-GRU Model," IEEE Access, vol. 10, pp. 39313–39324, 2022, doi: 10.1109/ACCESS.2022.3165621.

[22] Z. Gao, Z. Li, J. Luo, and X. Li, "Short Text Aspect-Based Sentiment Analysis Based on CNN + BiGRU," Appl. Sci., vol. 12, no. 5, p. 2707, Mar. 2022, doi: 10.3390/app12052707.

[23] J. Wang, Y. Zhang, L.-C. Yu, and X. Zhang, "Contextual sentiment embeddings via bi-directional GRU language model," Knowledge-Based Syst., vol. 235, p. 107663, Jan. 2022, doi: 10.1016/j.knosys.2021.107663.

[24] B. Gupta, P. Prakasam, and T. Velmurugan, "Integrated BERT embeddings, BiLSTM-BiGRU and 1-D CNN model for binary sentiment classification analysis of movie reviews," Multimed. Tools Appl., vol. 81, no. 23, pp. 33067–33086, Sep. 2022, doi: 10.1007/s11042-022-13155-w.

[25] A. Kenarang, M. Farahani, and M. Manthouri, "BiGRU attention capsule neural network for persian text classification," J. Ambient Intell. Humaniz. Comput., vol. 13, no. 8, pp. 3923–3933, Aug. 2022, doi: 10.1007/s12652-022-03742-y.

[26] H. Yanagimoto and K. Hashimoto, "GCNN With Self-Attention is Better Than GRU With Self-Attention For Sentiment Analysis," ICIC Express Lett., vol. 16, no. 5, pp. 497–504, 2022, doi: 10.24507/icicel.16.05.497.

[27] X. Zhang, Z. Wu, K. Liu, Z. Zhao, J. Wang, and C. Wu, "Text Sentiment Classification Based on BERT Embedding and Sliced Multi-Head Self-Attention Bi-GRU," Sensors, vol. 23, no. 3, p. 1481, Jan. 2023, doi: 10.3390/s23031481.

[28] M. S. Başarslan and F. Kayaalp, "MBi-GRUMCONV: A novel Multi Bi-GRU and Multi CNN-Based deep learning model for social media sentiment analysis," J. Cloud Comput., vol. 12, no. 1, p. 5, Jan. 2023, doi: 10.1186/s13677-022-00386-3.

[29] N. Habbat, H. Anoun, and L. Hassouni, "Combination of GRU and CNN Deep Learning Models for Sentiment Analysis on French Customer Reviews Using XLNet Model," IEEE Eng. Manag. Rev., vol. 51, no. 1, pp. 41–51, Mar. 2023, doi: 10.1109/EMR.2022.3208818.

[30] Y. Huang, X. Dai, J. Yu, and Z. Huang, "SA-SGRU: Combining Improved Self-Attention and Skip-GRU for Text Classification," Appl. Sci., vol. 13, no. 3, p. 1296, Jan. 2023, doi: 10.3390/app13031296.

[31] K. L. Tan, C. P. Lee, and K. M. Lim, "RoBERTa-GRU: A Hybrid Deep Learning Model for Enhanced Sentiment Analysis," Appl. Sci., vol. 13, no. 6, p. 3915, Mar. 2023, doi: 10.3390/app13063915.

[32] X. Li, Y. Zhang, J. Jin, F. Sun, N. Li, and S. Liang, "A model of integrating convolution and BiGRU dual-channel mechanism for Chinese medical text classifications," PLoS One, vol. 18, no. 3, p. e0282824, Mar. 2023, doi: 10.1371/journal.pone.0282824.

[33] B. Zhang and W. Zhou, "Transformer-Encoder-GRU (T-E-GRU) for Chinese Sentiment Analysis on Chinese Comment Text," Neural Process. Lett., vol. 55, no. 2, pp. 1847–1867, Apr. 2023, doi: 10.1007/s11063-022-10966-8.

[34] W. Wei and X. Zhao, "Fault text classification of on-board equipment in high-speed railway based on labeled-Doc2vec and BiGRU," J. Rail Transp. Plan. Manag., vol. 26, p. 100372, Jun. 2023, doi: 10.1016/j.jrtpm.2023.100372.

[35] Y. Qin, Y. Shi, X. Hao, and J. Liu, "Microblog Text Emotion Classification Algorithm Based on TCN-BiGRU and Dual Attention," Information, vol. 14, no. 2, p. 90, Feb. 2023, doi: 10.3390/info14020090.

[36] Z. Li, L. Zhou, X. Yang, H. Jia, W. Li, and J. Zhang, "User Sentiment Analysis of COVID-19 via Adversarial Training Based on the BERT-FGM-BiGRU Model," Systems, vol. 11, no. 3, p. 129, Mar. 2023, doi: 10.3390/systems11030129.

[37] Y. Liu, "Automatic Classification Method for Chinese Patent Texts Based on BERT-BiGRU," J. Zhengzhou Univ. - Nat. Sci., vol. 55, no. 2, pp. 33–40, 2023, doi: 10.13705/j.issn.1671-6841.2022125.

[38] J. Wu, X. Zheng, J. Wang, J. Wu, and J. Wang, "AB-GRU: An attention-based bidirectional GRU model for multimodal sentiment fusion and analysis," Math. Biosci. Eng., vol. 20, no. 10, pp. 18523–18544, 2023, doi: 10.3934/mbe.2023822.

[39] T. Han, Z. Zhang, M. Ren, C. Dong, X. Jiang, and Q. Zhuang, "Text Emotion Recognition Based on XLNet-BiGRU-Att," Electronics, vol. 12, no. 12, p. 2704, Jun. 2023, doi: 10.3390/electronics12122704.

[40] D. Viji and S. Revathy, "A hybrid approach of Poisson distribution LDA with deep Siamese Bi-LSTM and GRU model for semantic similarity prediction for text data," Multimed. Tools Appl., vol. 82, no. 24, pp. 37221–37248, Oct. 2023, doi: 10.1007/s11042-023-15050-4.

[41] X. Zhang, L. Yu, and S. Tian, "BGAT: Aspect-based sentiment analysis based on bidirectional GRU and graph attention network," J. Intell. Fuzzy Syst., vol. 44, no. 2, pp. 3115–3126, Jan. 2023, doi: 10.3233/JIFS-213020.

[42] A. Pradhan, "A multichannel embedding and arithmetic optimized stacked Bi-GRU model with semantic attention to detect emotion over text data," Appl. Intell., vol. 53, no. 7, pp. 7647–7664, 2023, doi: 10.1007/s10489-022-03907-4.

[43] T. Shi, P. Li, W. Yang, A. Qi, and J. Qiao, "Application of TCN-biGRU neural network in $$ PM_{2.5}$$concentration prediction," Environ. Sci. Pollut. Res., vol. 30, no. 56, pp. 119506–119517, 2023, doi: 10.1007/s11356-023-30354-6.

[44] A. S. Girsang and Stanley, "Hybrid LSTM and GRU for Cryptocurrency Price Forecasting Based on Social Network Sentiment Analysis Using FinBERT," IEEE Access, vol. 11, pp. 120530–120540, 2023, doi: 10.1109/ACCESS.2023.3324535.

[45] P. He, H. Qi, S. Wang, and J. Cang, "Cross-Modal Sentiment Analysis of Text and Video Based on Bi-GRU Cyclic Network and Correlation Enhancement," Appl. Sci., vol. 13, no. 13, 2023, doi: 10.3390/app13137489.

[46] Y. Xiu, X. Liu, J. Qiu, T. Qu, J. Liu, and Y. Bian, "A Method of Sentiment Analysis and Visualized Interaction Based on Ernie-Tiny and BiGRU," Appl. Sci., vol. 13, no. 10, 2023, doi: 10.3390/app13105961.

[47] S. Abdulateef Waheeb, "Discharge Summaries Based Sentiment Detection Using Multi-Head Attention and CNN-BiGRU," Comput. Syst. Sci. Eng., vol. 46, no. 1, pp. 981–998, 2023, doi: 10.32604/csse.2023.035753.

[48] Y. Salini, P. Eswaraiah, M. V Brahmam, and U. Sirisha, "Word Embedding for Text Classification: Efficient CNN and Bi-GRU Fusion Multi Attention Mechanism," EAI Endorsed Trans. Scalable Inf. Syst., vol. 10, 2023, doi: 10.4108/eetsis.3992.

[49] Q. Wang, J. Tian, M. Li, and M. Lu, "Text Classification Based on CNN-BiGRU and Its Application in Telephone Comments Recognition," Int. J. Comput. Intell. Appl., vol. 22, no. 04, Dec. 2023, doi: 10.1142/S1469026823500219.

[50] M. Zulqarnain, R. Ghazali, M. Aamir, and Y. M. M. Hassim, "An efficient two-state GRU based on feature attention mechanism for sentiment analysis," Multimed. Tools Appl., vol. 83, no. 1, pp. 3085–3110, Jan. 2024, doi: 10.1007/s11042-022-13339-4.

[51] Z. Wang, S. Hu, and W. Liu, "Product feature sentiment analysis based on GRU-CAP considering Chinese sarcasm recognition," Expert Syst. Appl., vol. 241, p. 122512, May 2024, doi: 10.1016/j.eswa.2023.122512.

[52] Y. Wang, L. Feng, A. Liu, W. Wang, and Y. Hou, "Dual BiGRU-CNN-based sentiment classification method combining global and local attention," J. Supercomput., vol. 80, no. 2, pp. 2799–2837, Jan. 2024, doi: 10.1007/s11227-023-05558-9.

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[53] D. Wu, Z. Wang, and W. Zhao, "XLNet-CNN-GRU dual-channel aspect-level review text sentiment classification method," Multimed. Tools Appl., vol. 83, no. 2, pp. 5871–5892, Jan. 2024, doi: 10.1007/s11042-023-15026-4.

[54] P. Eswaraiah and H. Syed, "A Hybrid Deep Learning GRU based Approach for Text Classification using Word Embedding," EAI Endorsed Trans. Internet Things, vol. 10, pp. 1–8, 2024, doi: 10.4108/eetiot.4590.

[55] S. Yu, D. Liu, W. Zhu, Y. Zhang, and S. Zhao, "Attention-based LSTM, GRU and CNN for short text classification," J. Intell. Fuzzy Syst., vol. 39, no. 1, pp. 333–340, Jul. 2020, doi: 10.3233/JIFS-191171.

[56] Y. Salini, P. Eswaraiah, M. V. Brahmam, and U. Sirisha, "Word Embedding for Text Classification: Efficient CNN and Bi-GRU Fusion Multi Attention Mechanism," ICST Trans. Scalable Inf. Syst., vol. 10, Sep. 2023, doi: 10.4108/eetsis.3992.

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